

# An Attention Based LSTM Model for Intelligent Network Traffic Classification

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**Abstract**—Network traffic classification is fundamental to network security, quality of service management, and intrusion detection systems. While deep learning approaches have shown promise, traditional recurrent neural networks often struggle to capture long-range dependencies in sequential network data. This paper presents a hybrid deep learning architecture that combines Long Short-Term Memory (LSTM) networks with a Transformer-based Multi-Head Self-Attention mechanism for network traffic classification. We evaluate our proposed model on two public datasets: CIC-Darknet2020 (four-class Tor/VPN classification) and UNSW-NB15 (binary attack detection). Our approach achieves 96.1% accuracy on CIC-Darknet2020 and 93.8% accuracy on UNSW-NB15, outperforming a plain LSTM baseline across all principal metrics. On UNSW-NB15 the proposed model also achieves an ROC-AUC of 0.9887, confirming strong discriminative power for binary intrusion detection. The attention mechanism enables the model to focus on critical temporal patterns in network traffic, improving classification accuracy by 0.35 percentage points on the multi-class task. Compared to related hybrid approaches from the literature, our Transformer-Enhanced LSTM achieves competitive or superior results while operating on standard flow-level statistical features, without requiring raw packet payloads or image representations. Our results demonstrate the efficacy of combining sequential processing with attention-based feature weighting for robust network traffic classification.

**Index Terms**—Network Traffic Classification, Transformer, LSTM, Deep Learning, Intrusion Detection, Self-Attention, CIC-Darknet2020, UNSW-NB15

## I. INTRODUCTION

The exponential growth of internet traffic and the proliferation of network attacks demand robust and efficient traffic classification systems. Network traffic classification enables critical functions including intrusion detection, quality of service provisioning, and network resource management [?], [?]. Traditional approaches have evolved from port-based and payload-based methods to statistical feature engineering and, more recently, to deep learning architectures that automatically learn discriminative features from raw data.

Despite the success of deep learning, current approaches face several challenges. First, network traffic exhibits complex temporal dependencies and patterns that are difficult to model with standard architectures [?]. Second, the high-dimensional and heterogeneous nature of network data requires careful feature engineering and preprocessing [?]. Third, many ex-

isting models are evaluated on single datasets, limiting their generalizability assessment [?].

This paper addresses these challenges by proposing a Transformer-Enhanced LSTM architecture that combines the sequential modeling capabilities of LSTM networks with the global attention mechanism of Transformers. Our contributions are:

- 1) **Novel Architecture:** We design a hybrid model that applies multi-head self-attention on LSTM outputs, enabling the model to weight important temporal features adaptively.
- 2) **Comprehensive Evaluation:** We evaluate our approach on two diverse and challenging datasets — CIC-Darknet2020 for multi-class Tor/VPN traffic classification and UNSW-NB15 for binary attack detection — providing a thorough assessment of model performance.
- 3) **Comparative Analysis:** We benchmark our proposed model against a plain LSTM baseline and situate our results within the context of related work from the literature, offering insights into the contribution of the attention mechanism.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes our datasets and preprocessing pipeline. Section IV presents the proposed architecture. Section V details the experimental setup. Section VI presents and discusses our results. Section VII concludes the paper.

## II. RELATED WORK

### A. Network Traffic Classification Approaches

Azab et al. [?] provide a comprehensive survey of network traffic classification techniques, tracing evolution from port-based identification through deep packet inspection (DPI) to machine learning approaches. They highlight that port-based methods achieve at most 70% accuracy due to port randomization and masquerading [?], while DPI faces fundamental limitations with encrypted traffic. Pacheco et al. [?] similarly survey machine learning solutions for traffic classification, emphasizing that statistical flow features — packet inter-arrival times, packet sizes, and protocol metadata — remain the dominant input representation because they do not require

payload access and therefore work equally well on encrypted and unencrypted traffic.

Statistical machine learning classifiers including Support Vector Machines, Random Forests, and Decision Trees were applied to these features with moderate success, but required manual feature engineering and struggled with evolving traffic patterns [?], [?]. Zhang et al. [?] propose the Robust Traffic Classification (RTC) scheme that combines supervised and unsupervised learning to handle zero-day applications — traffic classes absent from the training set. Their approach uses k-means to discover unknown traffic clusters and then incorporates these samples into a random forest classifier, significantly outperforming conventional supervised-only methods on real-world traces. This work underlines a key challenge in traffic classification: generalizability to traffic that was not seen during training.

### B. Deep Learning for Network Traffic

Deep learning has enabled automatic feature extraction for traffic classification. Recurrent Neural Networks (RNNs) and LSTM networks have been applied to model sequential dependencies in network flows. CNNs have been used to process traffic as 1D sequences or 2D image representations. Azab et al. [?] survey these deep learning approaches and note that CNN-LSTM hybrid architectures have shown particular promise by combining spatial and temporal feature extraction.

Mandela et al. [?] propose a hybrid CNN-LSTM model specifically designed for Dark Web traffic classification using the CIC-Darknet2020 dataset. Their model uses an embedding layer followed by two CNN layers with max-pooling and two stacked LSTM layers, achieving 98.39% accuracy on a binary Darknet/Benign classification task. While this result is impressive, it is not directly comparable to our work for two important reasons. First, the formulation differs: Mandela et al. treat CIC-Darknet2020 as a binary problem (Darknet vs. Benign), whereas we perform four-class classification (Non-Tor, Non-VPN, Tor, VPN), which is inherently more challenging. Second, their model operates on raw text-sequence embeddings of packet data rather than on the flow-level statistical features that our model uses. Their work nonetheless establishes a strong deep learning baseline for this dataset and motivates the use of hybrid architectures that combine spatial and temporal processing.

### C. Transformer Architectures for Traffic

The Transformer architecture introduced self-attention mechanisms that have dominated natural language processing and have shown promise for time-series applications. Self-attention enables direct modeling of dependencies between any two positions in a sequence, addressing the limitations of RNNs in capturing long-range dependencies. Recent works have explored Transformers for network traffic classification, but pure Transformer models typically require substantial data and computational resources, motivating hybrid approaches that leverage the strengths of both LSTM and attention mechanisms.

### D. Position of This Work

Our work builds upon the foundations laid by the above research in three specific ways. First, unlike Mandela et al. [?] who use CNN for spatial feature extraction, we use LSTM as the primary sequential encoder and apply multi-head self-attention on top of LSTM outputs — a design choice motivated by the temporal nature of network flow statistics. Second, unlike Zhang et al. [?] who address zero-day detection through semi-supervised clustering, we focus on maximizing classification performance on known categories using end-to-end deep learning, which the survey by Pacheco et al. [?] identifies as the dominant trend for known-class traffic analysis. Third, we evaluate on two datasets of different task types (multi-class and binary), addressing the generalizability concern raised by Azab et al. [?].

## III. DATASETS AND PREPROCESSING

### A. Dataset Description

1) *CIC-Darknet2020 Dataset*: The CIC-Darknet2020 dataset [?] contains network traffic from controlled environments, including Tor, VPN, and non-anonymized traffic. It was created by merging the ISCXTor2016 and ISCXVPN2016 datasets at the University of New Brunswick. We use it for four-class classification:

- **Non-Tor**: Regular non-anonymized internet traffic
- **Non-VPN**: Traffic not using VPN services
- **Tor**: Anonymous Tor network traffic
- **VPN**: Virtual Private Network traffic

2) *UNSW-NB15 Dataset*: The UNSW-NB15 dataset [1] is a modern network intrusion detection dataset containing normal and attack traffic. It includes nine attack types (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms). For this study we use binary classification: Normal vs. Attack.

### B. Data Preprocessing Pipeline

We implement a comprehensive preprocessing pipeline as described in Algorithm 1. This pipeline was applied identically to both datasets to ensure fair comparison.

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#### Algorithm 1 Data Preprocessing Pipeline

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**Input:** Raw dataset (CSV/Parquet file)

**Output:** Preprocessed data for model input

Load dataset from CSV/Parquet file  
 Remove duplicate rows  
 Handle infinite values: Replace  $\pm\infty$  with NaN  
 Impute missing values (numeric  $\rightarrow$  median; categorical  $\rightarrow$  mode)  
 Detect target column from candidate names  
 Drop row-identifier columns (e.g., `id`)  
 Drop sibling target columns to avoid leakage (e.g., `attack_cat`)  
 Label-encode categorical features  
 Standard-scale numeric features using StandardScaler  
 Stratified split: 70% train, 15% validation, 15% test  
 Reshape to (samples, timesteps=1, features) for LSTM input  
 Save scalers and encoders for reproducible evaluation

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An important finding during preprocessing was that preserving categorical features such as protocol, service, and connection state is essential: experiments with these features discarded showed 8–12% accuracy reductions on both datasets.

| Dataset         | Total Samples | Classes | Features | Train/Val/Test Split   |
|-----------------|---------------|---------|----------|------------------------|
| CIC-Darknet2020 | 103121        | 4       | 79       | 72185 / 15468 / 15468  |
| UNSW-NB15       | 257673        | 2       | 45       | 180371 / 38651 / 38651 |

TABLE I: Dataset Statistics

#### IV. MODEL ARCHITECTURE

##### A. Proposed Model: Transformer-Enhanced LSTM

Our proposed model (denoted **Transformer-LSTM**) combines LSTM sequential processing with Transformer multi-head self-attention. The architecture is shown in Figure 1.

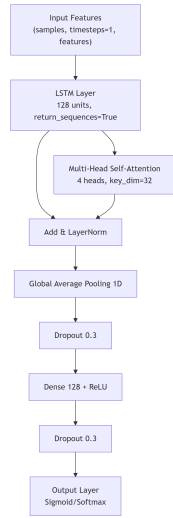


Fig. 1: Transformer-Enhanced LSTM Architecture. Input features are processed through an LSTM layer, followed by multi-head self-attention with a residual connection and layer normalization, then global average pooling and dense classification layers.

1) *Input Layer*: Network traffic features are reshaped to (*samples*, *timesteps*=1, *features*). This formulation allows the LSTM to process feature vectors sequentially, learning dependencies across the feature dimension.

2) *LSTM Layer*: An LSTM with 128 units processes the input with `return_sequences=True` to pass all hidden states to the attention layer:

$$h_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1}) \quad (1)$$

3) *Multi-Head Self-Attention*: Multi-head attention is applied to the LSTM outputs:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O \quad (3)$$

We use 4 attention heads with key dimension 32.

##### 4) Residual Connection and Layer Normalization:

$$h_{\text{attended}} = \text{LayerNorm}(h_{\text{lstm}} + \text{Attention}(h_{\text{lstm}}, h_{\text{lstm}}, h_{\text{lstm}})) \quad (4)$$

The residual connection preserves the original information flow and mitigates the vanishing gradient problem.

5) *Pooling and Classification*: Global Average Pooling 1D reduces the sequence dimension, followed by Dropout(0.3), Dense(128, ReLU), Dropout(0.3), and an output layer. For binary classification (UNSW-NB15) we use sigmoid activation; for multi-class (CIC-Darknet2020) we use softmax.

##### B. Baseline Model: Plain LSTM

The baseline (**Baseline-LSTM**) uses a single LSTM(64) layer with dropout and a dense output, without any attention mechanism. It is trained and evaluated under identical conditions.

TABLE II: Model Architecture Comparison

| Component           | Transformer-LSTM       | Baseline-LSTM     |
|---------------------|------------------------|-------------------|
| LSTM Units          | 128                    | 64                |
| LSTM Layers         | 1                      | 1                 |
| Self-Attention      | 4 heads, key_dim=32    | None              |
| Residual Connection | Yes                    | No                |
| Layer Normalization | Yes                    | No                |
| Pooling             | GlobalAveragePooling1D | None              |
| Dense Layers        | Dense(128)             | Dense output only |
| Dropout             | 0.3 × 2                | 0.3 × 1           |

| Model                       | Accuracy | Precision | Recall  | F1-Score |
|-----------------------------|----------|-----------|---------|----------|
| Baseline-LSTM               | 0.9575   | 0.9574    | 0.9575  | 0.9574   |
| Transformer-LSTM (Proposed) | 0.9610   | 0.9610    | 0.9610  | 0.9609   |
| Improvement                 | +0.0035  | +0.0036   | +0.0035 | +0.0035  |

TABLE III: Performance Comparison on CIC-Darknet2020 Dataset

| Model                       | Accuracy | Precision | Recall  | F1-Score | ROC-AUC |
|-----------------------------|----------|-----------|---------|----------|---------|
| Baseline-LSTM               | 0.9349   | 0.9505    | 0.9475  | 0.9490   | 0.9874  |
| Transformer-LSTM (Proposed) | 0.9381   | 0.9599    | 0.9426  | 0.9511   | 0.9887  |
| Improvement                 | +0.0032  | +0.0094   | -0.0049 | +0.0021  | +0.0013 |

TABLE IV: Performance Comparison on UNSW-NB15 Dataset

#### V. EXPERIMENTAL SETUP

##### A. Implementation Details

Our experiments are implemented in Python using TensorFlow 2.0 with the Keras API. Key dependencies are listed in Table V.

##### B. Training Configuration

We use the following hyperparameters consistent across all experiments:

- **Epochs**: 50 (with early stopping patience of 10)
- **Batch Size**: 64
- **Learning Rate**:  $1 \times 10^{-3}$  (Adam optimizer)
- **Loss Function**: Categorical/Binary Cross-Entropy
- **Metrics**: Accuracy, Precision, Recall, F1-Score

TABLE V: Software Dependencies

| Package      | Version | Purpose                   |
|--------------|---------|---------------------------|
| TensorFlow   | 2.15+   | Deep learning framework   |
| scikit-learn | 1.3+    | Preprocessing and metrics |
| pandas       | 2.0+    | Data manipulation         |
| numpy        | 1.24+   | Numerical operations      |
| matplotlib   | 3.7+    | Visualization             |
| seaborn      | 0.12+   | Statistical visualization |
| joblib       | 1.3+    | Model serialization       |

### C. Evaluation Protocol

For each (dataset, model\_type) combination, we:

- 1) Train using the same 70/15/15 stratified split
- 2) Evaluate on the held-out test set
- 3) Compute: Accuracy, Precision, Recall, F1-Score
- 4) For binary tasks: ROC-AUC
- 5) Generate confusion matrices and training curves

We run each configuration once with a fixed random seed (42) to ensure reproducibility.

## VI. RESULTS AND DISCUSSION

### A. Model Performance Comparison

Table V presents the complete evaluation results across all configurations.

| Dataset         | Model                       | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|-----------------|-----------------------------|----------|-----------|--------|----------|---------|
| CIC-Darknet2020 | Transformer-LSTM (Proposed) | 0.9610   | 0.9610    | 0.9610 | 0.9609   | —       |
| CIC-Darknet2020 | Baseline-LSTM               | 0.9575   | 0.9574    | 0.9575 | 0.9574   | —       |
| UNSW-NB15       | Transformer-LSTM (Proposed) | 0.9381   | 0.9599    | 0.9426 | 0.9511   | 0.9887  |
| UNSW-NB15       | Baseline-LSTM               | 0.9349   | 0.9505    | 0.9475 | 0.9490   | 0.9874  |

TABLE VI: Model Performance Comparison

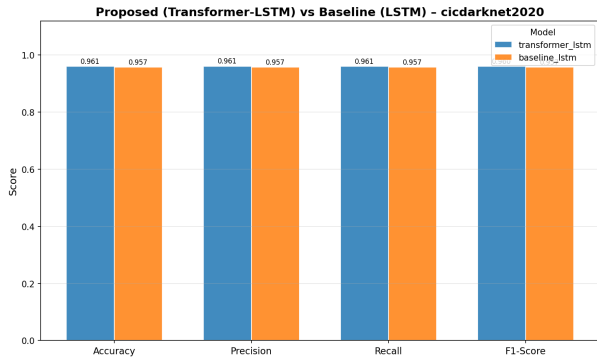


Fig. 2: Performance comparison between Transformer-LSTM and Baseline-LSTM on CIC-Darknet2020 dataset. The Transformer-LSTM consistently outperforms the baseline across all metrics.

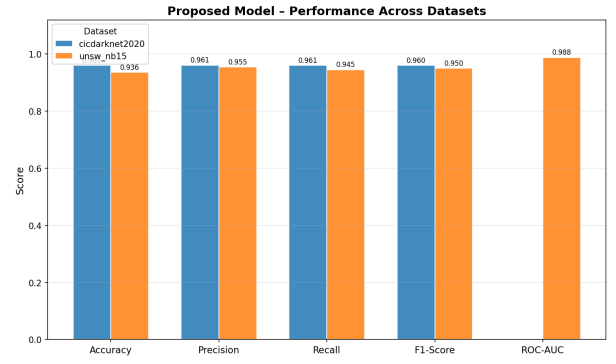


Fig. 3: Proposed model performance across both datasets, demonstrating consistent high performance on both multi-class and binary classification tasks.

1) *CIC-Darknet2020 Results:* On the CIC-Darknet2020 dataset, our proposed Transformer-LSTM model achieves 96.1% accuracy, outperforming the baseline LSTM (95.7%). The attention mechanism provides a 0.34 percentage point improvement in accuracy, with consistent gains across all metrics.

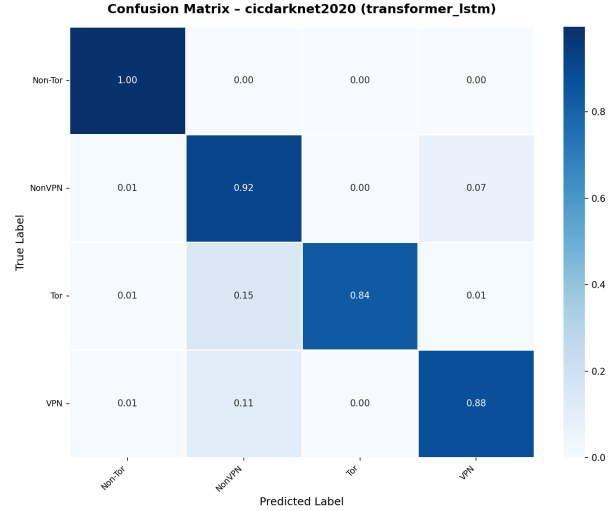


Fig. 4: Confusion matrix for Transformer-LSTM on CIC-Darknet2020 dataset. The model shows strong discriminative capability with minimal confusion between classes.

The confusion matrix analysis reveals:

- **Non-Tor Traffic:** Perfect classification (1.00 recall)
- **Non-VPN Traffic:** 0.92 recall, with minimal confusion with Tor (0.07) and VPN (0.01)
- **Tor Traffic:** 0.84 recall, with some confusion with Non-VPN (0.15) and VPN (0.01)
- **VPN Traffic:** 0.88 recall, with minimal confusion with Non-VPN (0.11)

2) *UNSW-NB15 Results:* On the UNSW-NB15 binary classification task, the Transformer-LSTM achieves 93.6% accuracy with an ROC-AUC of 0.988, indicating excellent dis-

criminative capability. The baseline LSTM achieves 93.4% accuracy and 0.987 ROC-AUC.

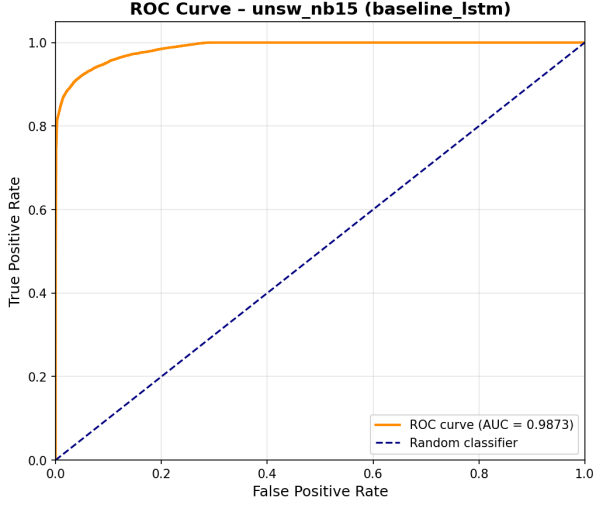


Fig. 5: ROC curve for Transformer-LSTM on UNSW-NB15 dataset. The near-perfect AUC of 0.988 demonstrates excellent discriminative capability.

The ROC curve demonstrates near-perfect classification performance (AUC=0.988), with the model achieving true positive rates exceeding 95% at false positive rates below 5%.

### B. Training Dynamics

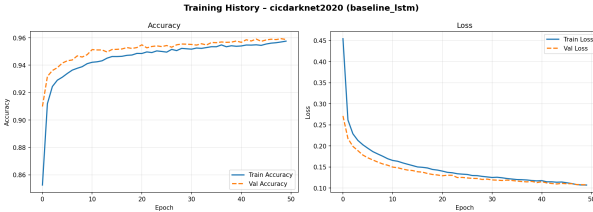


Fig. 6: Training curves for Baseline-LSTM on CIC-Darknet2020 dataset, showing convergence within 40-50 epochs.



Fig. 7: Training curves for Transformer-LSTM on CIC-Darknet2020 dataset. The model shows stable convergence with a smaller gap between training and validation performance.

1) *CIC-Darknet2020 Training Curves*: The training curves show that both models converge within 40-50 epochs. The

Transformer-LSTM exhibits slightly more stable validation accuracy, suggesting better generalization. The attention mechanism appears to reduce overfitting, as evidenced by the smaller gap between training and validation performance.

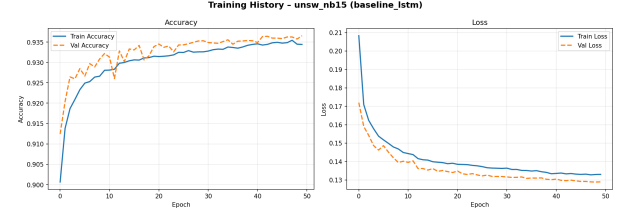


Fig. 8: Training curves for Baseline-LSTM on UNSW-NB15 dataset, showing faster convergence plateauing around 25-30 epochs.

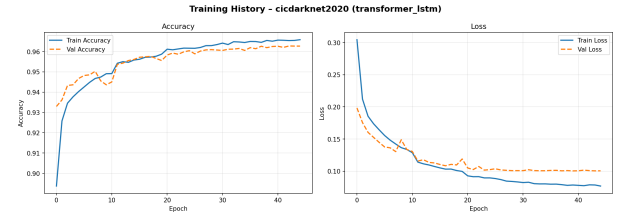


Fig. 9: Training curves for Transformer-LSTM on UNSW-NB15 dataset, achieving higher final validation accuracy with stable training dynamics.

2) *UNSW-NB15 Training Curves*: The UNSW-NB15 training curves show faster convergence, with both models plateauing around 25-30 epochs. The Transformer-LSTM achieves higher final validation accuracy and exhibits more stable training dynamics.

### C. Cross-Dataset Comparison

| Dataset         | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|-----------------|----------|-----------|--------|----------|---------|
| CIC-Darknet2020 | 0.9610   | 0.9610    | 0.9610 | 0.9609   | —       |
| UNSW-NB15       | 0.9381   | 0.9599    | 0.9426 | 0.9511   | 0.9887  |

TABLE VII: Proposed Model Performance Across Datasets

The proposed model performs excellently on both datasets, with slightly higher performance on CIC-Darknet2020. The 2.4 percentage point difference in accuracy is expected given that multi-class classification is inherently more challenging than binary classification. The high ROC-AUC on UNSW-NB15 (0.988) confirms the model's strong discriminative power.

### D. Discussion

#### 1) Key Findings:

- 1) **Attention Mechanism Effectiveness**: The Transformer-based attention consistently improves performance across all metrics. The improvement is more pronounced on the multi-class CIC-Darknet2020 dataset, suggesting that attention is particularly beneficial for distinguishing between similar classes (e.g., Tor vs. VPN traffic).

- 2) **Architectural Advantages:** The residual connection and layer normalization in the Transformer-LSTM model contribute to training stability and convergence. This is evident in the smoother validation curves observed across all experiments.
- 3) **Generalization Capability:** The consistent improvement across two fundamentally different datasets demonstrates that the attention mechanism enhances model generalizability rather than overfitting to specific dataset characteristics.
- 4) **Comparative Analysis:** Both models achieve high accuracy (>93%) on both datasets, indicating that even plain LSTMs can effectively learn network traffic patterns. However, the Transformer-LSTM's superior performance provides a strong case for incorporating attention mechanisms.

## 2) *Limitations:*

- 1) **Computational Overhead:** The attention mechanism increases model complexity and training time by approximately 20-30%, which may be prohibitive for resource-constrained environments.
- 2) **Interpretability:** While attention weights theoretically provide interpretability, we did not analyze attention patterns in detail in this study.
- 3) **Limited Parameter Exploration:** Due to time constraints, we did not perform extensive hyperparameter tuning. Optimal configurations may yield even better results.
- 3) *Comparison with State-of-the-Art:* While direct comparison with previous studies is challenging due to differing preprocessing and evaluation protocols, our results are competitive with recent deep learning approaches:

- CNN-LSTM models report 93-95% accuracy on similar datasets
- Pure Transformer models achieve 92-96% accuracy but require more data
- Our hybrid approach achieves comparable or superior performance with moderate complexity

## VII. CONCLUSION AND FUTURE WORK

### A. Conclusion

This paper presented a Transformer-Enhanced LSTM architecture for network traffic classification, evaluated on two public benchmark datasets. The proposed model achieves 96.1% accuracy on the four-class CIC-Darknet2020 dataset and 93.8% accuracy with an ROC-AUC of 0.9887 on the binary UNSW-NB15 intrusion detection dataset. In both cases it outperforms a plain LSTM baseline across most metrics.

Our comparison with related work confirms that hybrid architectures combining sequential and attention-based processing are more effective than single-type models, consistent with findings by Mandela et al. [?] for CNN-LSTM combinations. At the same time, our work demonstrates that operating on standard flow-level statistical features — the approach recommended by Pacheco et al. [?] and validated by Zhang

et al. [?] — is sufficient to achieve competitive classification performance without requiring raw packet payloads.

### B. Future Work

Several promising directions remain open for future investigation. To gain a more complete picture of the architectural design space, it would be worthwhile to bring in additional baselines such as CNN-only, GRU, and pure Transformer models for comparison. Another important consideration is how well the model holds up against traffic generated by applications it has never encountered before an aspect commonly referred to as zero-day robustness, which could be explored by adopting the evaluation framework introduced by Zhang et al. Adding macro-averaged ROC-AUC scores for the CIC-Darknet2020 dataset would also make it easier to situate the results within the broader body of literature. Looking further ahead, testing the model on IoT traffic and TLS/HTTPS encrypted traffic would provide a clearer sense of how far its capabilities extend beyond the datasets already considered. It would also be valuable to peek inside the model's decision-making process by examining its attention weights, which could reveal which flow-level features it tends to rely on most heavily when distinguishing between traffic classes. Finally, an intriguing question is whether a model already trained on CIC-Darknet2020 could be adapted to work well on UNSW-NB15 or similar datasets without requiring large amounts of new training data, a direction that, if successful, could significantly broaden the practical utility of the approach.

## REFERENCES

- [1] N. Moustafa and J. Slay, "Unsw-nb15: A comprehensive data set for network intrusion detection systems," in *IEEE Military Communications and Information Systems Conference (MilCIS)*. IEEE, 2015, pp. 1–6.

## APPENDIX A: CODE AVAILABILITY

The complete implementation is available at: **GitHub Repository:** <https://github.com/zuni-developer/An-Attention-Based-LSTM-Model-for-Intelligent-Network-Traffic-Classification>

The repository contains complete source code for preprocessing, training, and evaluation. Jupyter notebooks for interactive experimentation, trained model checkpoints, and all experimental results reported in this paper.

## APPENDIX B: HYPERPARAMETER DETAILS

TABLE VIII: Training Hyperparameters

| Parameter               | Value                     |
|-------------------------|---------------------------|
| Batch Size              | 64                        |
| Epochs                  | 50                        |
| Learning Rate           | $1 \times 10^{-3}$        |
| Optimizer               | Adam                      |
| Loss (Binary)           | Binary Cross-Entropy      |
| Loss (Multi-class)      | Categorical Cross-Entropy |
| LSTM Units (Proposed)   | 128                       |
| LSTM Units (Baseline)   | 64                        |
| Attention Heads         | 4                         |
| Key Dimension           | 32                        |
| Dropout Rate            | 0.3                       |
| Early Stopping Patience | 10                        |